

```
In [1]: import pandas as pd
        from sklearn.model_selection import train_test_split
        from sklearn.linear_model import LogisticRegression
        from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score
```

```
In [6]: df = pd.read_csv(r"D:\Pradnya\TE\DSBDA\Practicals\new\sociaInetwork\Social_Network.csv")
        print(df.head())
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

```
In [11]: X = df[['Age', 'EstimatedSalary']]
        y = df['Purchased']
        X_train, X_test, y_train, y_test = train_test_split(
            X, y, test_size=0.2, random_state=42)
        model = LogisticRegression()
        model.fit(X_train, y_train)

        y_pred = model.predict(X_test)
```

```
In [8]: cm = confusion_matrix(y_test, y_pred)
        print("\nConfusion Matrix:\n", cm)
```

Confusion Matrix:

```
[[50  2]
 [ 7 21]]
```

```
In [9]: TN, FP, FN, TP = cm.ravel()
        print("\nTN:", TN)
        print("FP:", FP)
        print("FN:", FN)
        print("TP:", TP)
```

TN: 50

FP: 2

FN: 7

TP: 21

```
In [10]: print("\nAccuracy:", accuracy_score(y_test, y_pred))
        print("Error Rate:", 1 - accuracy_score(y_test, y_pred))
        print("Precision:", precision_score(y_test, y_pred))
        print("Recall:", recall_score(y_test, y_pred))
```

Accuracy: 0.8875

Error Rate: 0.11250000000000004

Precision: 0.9130434782608695

Recall: 0.75

```
In [ ]:
```